

A Comprehensive Measure of Well-Being: Intelligent HDI Prediction System

Project Description:

The **Human Development Index (HDI) Prediction System** is a Machine Learning-based web application developed to predict the Human Development Index of a country using important socio-economic indicators. The system uses a trained **Linear Regression** model to estimate the HDI score based on **Life Expectancy, Mean Years of Schooling, Expected Years of Schooling, and Gross National Income (GNI) per Capita**.

The application is developed using **Python, Flask, HTML, CSS, and JavaScript**, providing a simple and user-friendly interface. Users can enter the required values through the web application, and the trained model instantly predicts the HDI score along with the corresponding development category (**Low, Medium, High, or Very High Human Development**).

The project helps researchers, students, and policymakers quickly estimate human development without performing manual calculations. It demonstrates how Machine Learning can support data-driven analysis and decision-making in socio-economic studies.

Scenarios

Scenario 1: A researcher enters the socio-economic indicators of a country with high life expectancy, better education levels, and high GNI per capita. The system predicts a **Very High Human Development** category with an appropriate HDI score.

Scenario 2: A student enters moderate values for life expectancy, education, and income. The system predicts a **Medium Human Development** category, helping the user understand the country's development level.

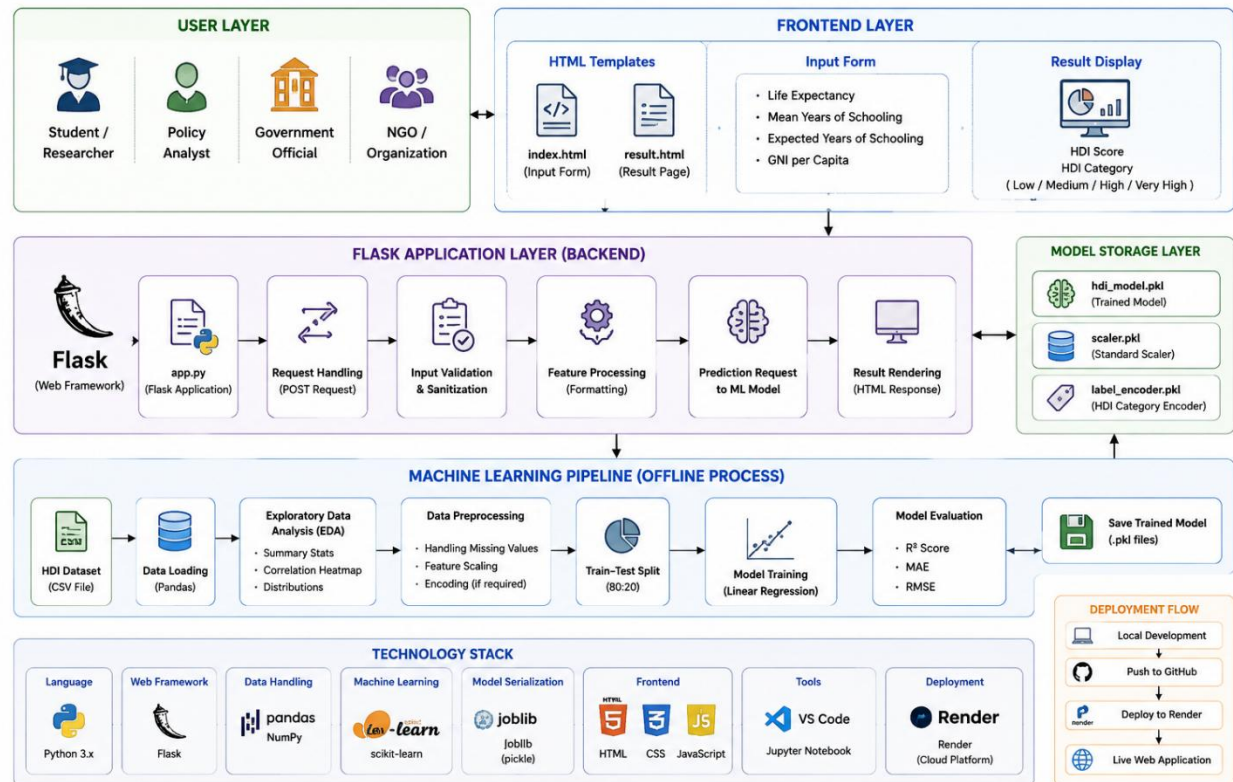
Scenario 3: A policymaker enters lower values for the development indicators of a country. The application predicts a **Low Human Development** category, assisting in identifying areas that require improvement.

Technical Architecture:

Description: The **Human Development Index (HDI) Prediction System** follows a **three-tier architecture** to provide fast and accurate HDI prediction. The **frontend** is developed using **HTML, CSS, and JavaScript**, where users enter the required socio-economic indicators. The **Flask backend** validates the user input, processes the data, and communicates with the trained **Linear Regression** model. The model predicts the **HDI score** and returns the result to the frontend, where the predicted score and corresponding **HDI category** are displayed.

The application is executed in the local development environment using the Flask framework. The prediction results are displayed through a web interface running on the local server.

TECHNICAL ARCHITECTURE – HDI PREDICTION SYSTEM



Pre-requisites:

- Python Programming Knowledge
- Flask Framework Knowledge
- Machine Learning Basics
- Linear Regression Concepts
- Pandas and NumPy
- Scikit-learn Library
- HTML, CSS, and JavaScript
- VS Code or Jupyter Notebook
- Git and GitHub Basics

Project Workflow:

Milestone 1: Dataset Preparation and System Architecture

- **Activity 1.1:** Collect and analyze the Human Development Index (HDI) dataset.
- **Activity 1.2:** Select the Linear Regression algorithm and prepare the dataset.
- **Activity 1.3:** Design the system architecture.

Milestone 2: Machine Learning Model Development

- **Activity 2.1:** Train and evaluate the Linear Regression model.
- **Activity 2.2:** Save the trained model using Pickle (.pkl).

Milestone 3: Flask Application Development

- **Activity 3.1:** Develop the Flask backend to load the trained model.
- **Activity 3.2:** Implement the HDI prediction logic and validate user inputs.

Milestone 4: Frontend Development

- **Activity 4.1:** Design the web interface using HTML, CSS, and JavaScript.
- **Activity 4.2:** The frontend was integrated with the Flask backend to submit user inputs and display the predicted HDI score, development category, and input summary on the result page.

Milestone 5: Local Testing

- **Activity 5.1:** Test the application using different input values.
- **Activity 5.2:** Verify prediction accuracy and fix identified issues.

Milestone 6: Conclusion

The HDI Prediction System uses a **Linear Regression-based Machine Learning model** and Flask to predict the Human Development Index from socio-economic indicators. The project includes data preprocessing, model training, backend and frontend development, and local testing to provide accurate HDI predictions through a simple web interface.

- Dataset preprocessed
- Training dataset ready

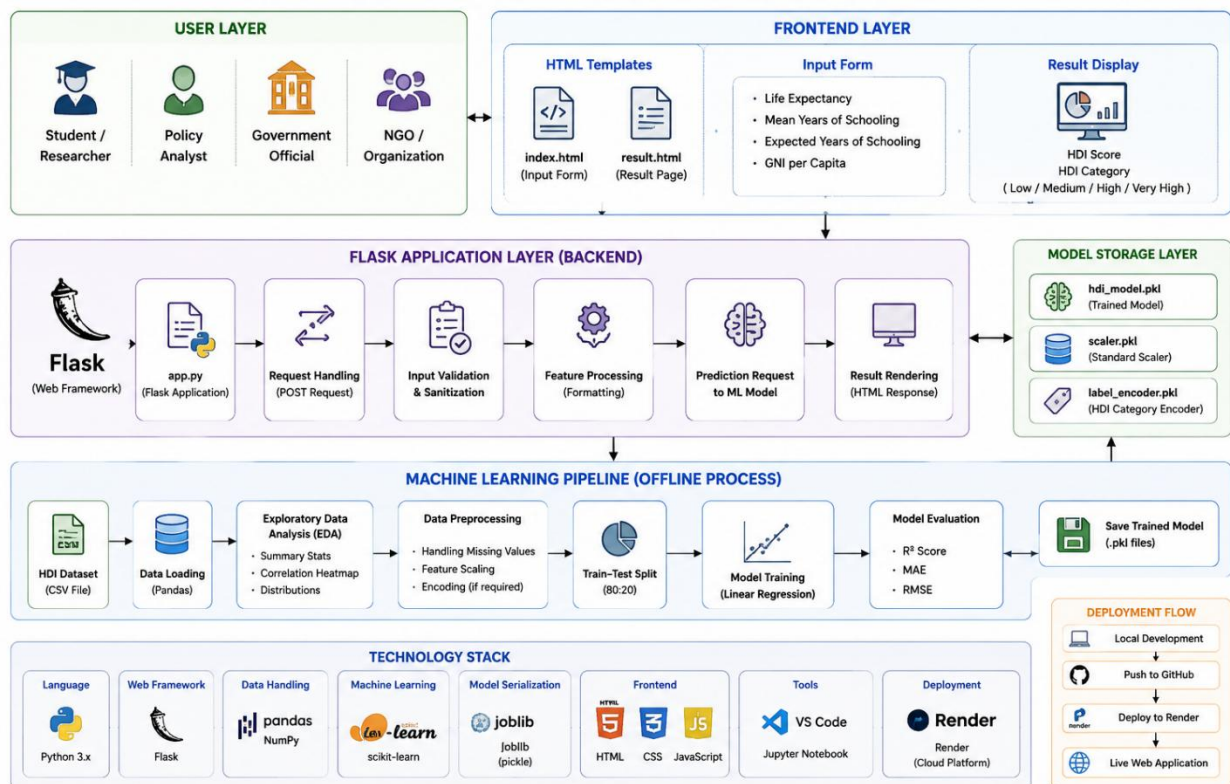
Activity 1.3: System Architecture Design

The technical architecture of the HDI Prediction System was designed. It explains how user inputs are processed through the Flask application, passed to the trained model, and displayed on the result page.

Output:

- System architecture completed
- Project workflow finalized

TECHNICAL ARCHITECTURE – HDI PREDICTION SYSTEM



Milestone 2: Machine Learning Model Development

This milestone focuses on developing the Machine Learning model for HDI prediction. The Linear Regression algorithm was trained using the prepared dataset, and the trained model was saved for future predictions.

Activity 2.1: Model Training

The Linear Regression model was trained using the processed HDI dataset. The model learned the relationship between the input indicators and the Human Development Index.

Output:

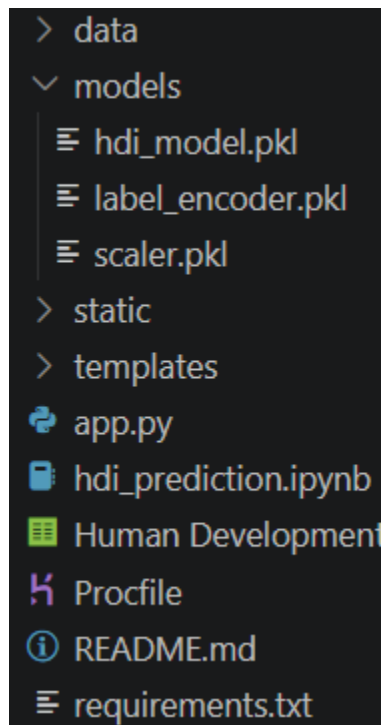
- Linear Regression model trained
- Prediction model generated

Activity 2.2: Model Saving

The trained model was saved as a .pkl file using the Pickle library. This model is later loaded by the Flask application to make predictions.

Output:

- Model saved successfully
- Prediction model ready for use



Milestone 3: Flask Application Development

This milestone focuses on developing the Flask backend application. The backend loads the trained model, receives user inputs, processes the data, and returns the predicted HDI score

Activity 3.1: Flask Backend Development

A Flask application was developed to load the trained model and handle prediction requests from the web interface.

Output:

- Flask backend created
- Model loaded successfully

Activity 3.2: Prediction Logic Implementation

Prediction logic was implemented to process user input, generate the HDI score using the trained model, and display the result.

Output:

- Prediction generated successfully
- HDI score returned to the user

```

app.py > get_hdi_category
1  from flask import Flask, render_template, request
2  import pandas as pd
3  import numpy as np
4  import joblib
5
6  app = Flask(__name__)
7
8  # Load Resources
9  def load_resources():
10     return (
11         joblib.load("models/hdi_model.pkl"),
12         joblib.load("models/scaler.pkl"),
13         joblib.load("models/label_encoder.pkl")
14     )
15
16  model, scaler, encoder = load_resources()
17
18
19  # Utility Functions
20  def get_hdi_category(score):
21     categories = [
22         (0.800, "Very High"),
23         (0.700, "High"),
24         (0.550, "Medium"),
25         (0.000, "Low")
26     ]
27
28     for limit, category in categories:
29         if score >= limit:
30             return category
31
32
33  def category_details(category):
34     details = {
35         "Very High": (
36             "very-high",
37             "🌟",
38             "This country ranks among the most developed nations with strong health, education, and income indicators."
39         ),
40         "High": (
41             "high",
42             "✅",
43             "This country demonstrates strong human development with good performance across most indicators."
44         ),
45         "Medium": (
46             "medium",
47             "📊",
48             "This country is developing. Improvements in health, education, and income can increase its HDI."
49         ),

```


app.py >  get_hdi_category

```
33 def category_details(category):
50     "Low": (
51         "low",
52         "📉",
53         "This country faces major development challenges in health, education, and economic growth."
54     )
55 }
56
57 color, emoji, desc = details[category]
58 return {
59     "color": color,
60     "emoji": emoji,
61     "description": desc
62 }
63
64
65 def validate(values):
66     limits = {
67         "Life Expectancy": (values[0], 20, 90),
68         "Mean Years of Schooling": (values[1], 0, 20),
69         "Expected Years of Schooling": (values[2], 0, 25),
70         "GNI Per Capita": (values[3], 100, 200000)
71     }
72
73     for name, (value, minimum, maximum) in limits.items():
74         if not minimum <= value <= maximum:
75             raise ValueError(f"{name} must be between {minimum} and {maximum}.")
76
77
78 def predict_hdi(values):
79     columns = [
80         "Life_Expectancy",
81         "Mean_Years_Schooling",
82         "Expected_Years_Schooling",
83         "GNI_Per_Capita"
84     ]
85
86     df = pd.DataFrame([values], columns=columns)
87
88     scaled = scaler.transform(df)
89     prediction = model.predict(scaled)[0]
90
91     return round(float(np.clip(prediction, 0, 1)), 3)
92
```

```
app.py > get_hdi_category
95     @app.route("/")
96     def home():
97         return render_template("index.html")
98
99
100    @app.route("/predict", methods=["POST"])
101    def predict():
102
103        try:
104            values = [
105                float(request.form["life_expectancy"]),
106                float(request.form["mean_years_schooling"]),
107                float(request.form["expected_years_schooling"]),
108                float(request.form["gni_per_capita"])
109            ]
110
111            validate(values)
112
113            score = predict_hdi(values)
114
115            category = get_hdi_category(score)
116
117            info = category_details(category)
118
119            return render_template(
120                "result.html",
121                hdi_score=score,
122                tier=category,
123                color=info["color"],
124                emoji=info["emoji"],
125                description=info["description"],
126                life_expectancy=values[0],
127                mean_years_schooling=values[1],
128                expected_years_schooling=values[2],
129                gni_per_capita=values[3]
130            )
131
132        except ValueError as error:
133            return render_template("index.html", error=str(error))
134
135        except Exception:
136            return render_template(
137                "index.html",
138                error="Something went wrong. Please check your inputs."
139            )
140
141
142    if __name__ == "__main__":
143        app.run(debug=True)
```

Milestone 4: Frontend Development

This milestone focuses on developing a simple and user-friendly web interface for the HDI Prediction System. The frontend was designed using HTML, CSS, and JavaScript to collect user inputs and display prediction results.

Activity 4.1: Designing the User Interface

The user interface was developed using HTML and CSS. Input fields were created for Life Expectancy, Mean Years of Schooling, Expected Years of Schooling, and GNI per Capita. The interface was designed to be clean, responsive, and easy to use.

Output:

- User interface developed
- Input form created successfully

```
69
70     <form action="/predict" method="POST">
71
72         <div class="input-grid">
73
74             <div class="input-group">
75
76                 <label> 📅 Life Expectancy</label>
77
78                 <input
79                     type="number"
80                     name="life_expectancy"
81                     step="0.1"
82                     min="20"
83                     max="90"
84                     placeholder="72.5"
85                     required>
86
87             </div>
88
89             <div class="input-group">
90
91                 <label> 📖 Mean Years of Schooling</label>
92
93                 <input
94                     type="number"
95                     name="mean_years_schooling"
96                     step="0.1"
97                     placeholder="8.4"
98                     required>
99
100             </div>
101
```

```

101
102     <div class="input-group">
103         <label> 📖 Expected Years of Schooling</label>
104
105         <input
106             type="number"
107             name="expected_years_schooling"
108             step="0.1"
109             placeholder="13.2"
110             required>
111
112     </div>
113
114     <div class="input-group">
115         <label> 💰 GNI Per Capita (PPP $)</label>
116
117         <input
118             type="number"
119             name="gni_per_capita"
120             placeholder="15000"
121             required>
122
123     </div>
124
125 </div>
126
127 <button type="submit" class="predict-btn">
128     Predict Human Development Index
129 </button>
130
131 </form>
132
133

```

Activity 4.2: Displaying Prediction Results

The frontend was connected to the Flask backend using JavaScript. After the user submits the input values, the predicted HDI score, development category, and input summary are displayed on the result page.

Output:

- Prediction results displayed successfully
- User-friendly result page implemented

```

35 <div class="result-card {{ color }}-result">
36
37   <div class="result-icon">|
38     {{ emoji }}
39   </div>
40
41   <h2 class="result-title">
42     {{ tier }} Human Development
43   </h2>
44
45   <p class="result-subtitle">
46     Human Development Index Prediction
47   </p>
48
49   <div class="score-box">
50
51     <span class="score-label">
52       HDI Score
53     </span>
54
55     <h1 class="score">
56       {{ hdi_score }}
57     </h1>
58
59   </div>
60
61   <div class="progress">
62     <div class="progress-fill"
63       | style="width: {{ (hdi_score * 100)|round|int }}%;">
64     </div>
65   </div>
66
67   <div class="progress-scale">
68     <span>0.0</span>
69     <span>0.55</span>
70     <span>0.70</span>
71     <span>0.80</span>
72     <span>1.0</span>
73   </div>
74   <p class="result-description">
75     {{ description }}
76   </p>
77
78 </div>
79

```

Milestone 5: Local Testing

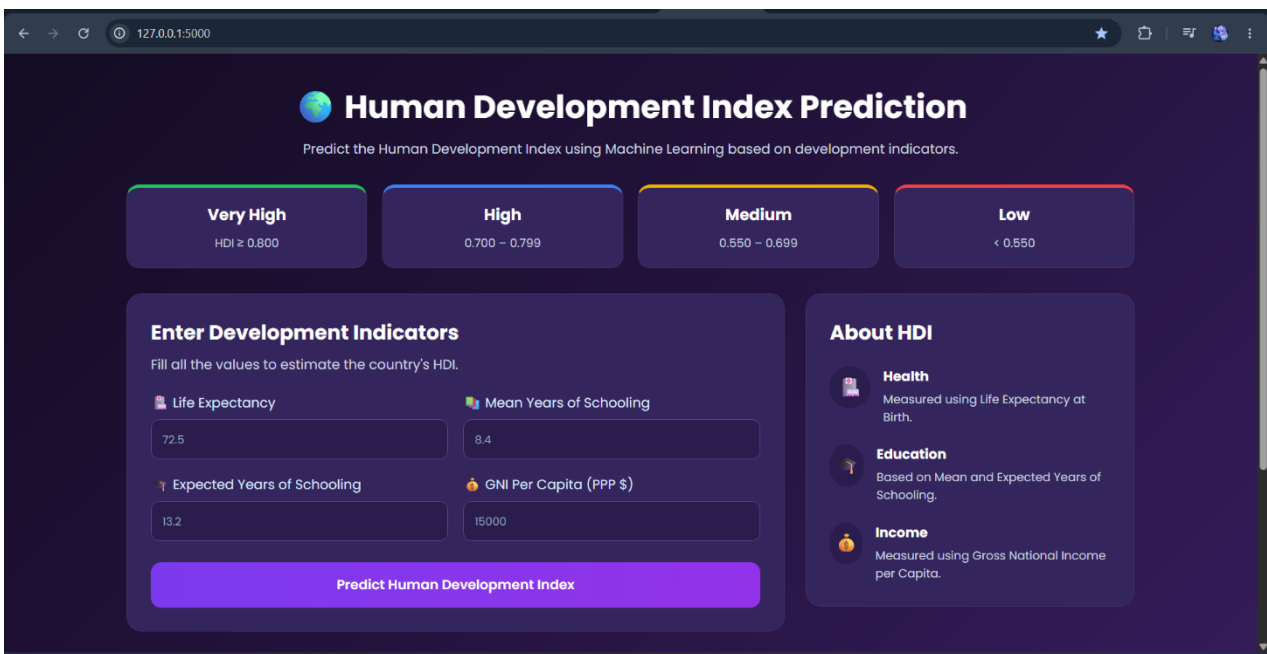
This milestone focuses on testing the HDI Prediction System in the local development environment. Different input values were used to verify that the application generates accurate HDI predictions and displays the results correctly.

Activity 5.1: Local Application Testing

The Flask application was executed on the local server, and multiple test cases were performed using different socio-economic input values. The prediction results were verified to ensure the application functioned correctly.

Output:

- Application tested successfully
- Predictions generated correctly



The screenshot shows a web browser window with the URL 127.0.0.1:5000. The application is titled "Human Development Index Prediction" and includes a subtitle: "Predict the Human Development Index using Machine Learning based on development indicators." The interface features four categories of HDI scores: "Very High" (HDI ≥ 0.800), "High" (0.700 - 0.799), "Medium" (0.550 - 0.699), and "Low" (< 0.550). Below these, there is a section titled "Enter Development Indicators" with a prompt: "Fill all the values to estimate the country's HDI." This section contains four input fields: "Life Expectancy" (72.5), "Mean Years of Schooling" (8.4), "Expected Years of Schooling" (13.2), and "GNI Per Capita (PPP \$)" (15000). A large purple button labeled "Predict Human Development Index" is positioned below the input fields. To the right, an "About HDI" section provides definitions for the indicators: "Health" (Measured using Life Expectancy at Birth), "Education" (Based on Mean and Expected Years of Schooling), and "Income" (Measured using Gross National Income per Capita).

Activity 5.2: Result Verification

The predicted HDI score and development category were verified using different input values. The user interface and prediction results were checked to ensure they were displayed correctly.

Output:

- Prediction results verified

- User interface tested successfully



Conclusion:

The HDI Prediction System successfully predicts the Human Development Index using a Linear Regression model based on key socio-economic indicators. The project integrates Machine Learning with Flask to provide a simple and user-friendly web application. It demonstrates how predictive analytics can support understanding of human development through quick and reliable HDI estimation.